



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025 26)
Faculty Name	Mohd. Razia Alangir Banu
Student Name	p yaswanth surya
Roll Number	2520030581
Branch	CSE
Course Outcome	CO-3

Case Study Topic: SmartGrid – Infrastructure Design & Network Optimization System

Work Done Summary:

The SmartGrid system was designed by modeling infrastructure as a weighted undirected graph, where nodes represent locations and edges represent connections with associated costs. BFS was implemented to traverse the network level by level and identify all reachable nodes, while DFS was used to explore deep paths and verify connectivity between different regions. To optimize infrastructure cost, Kruskal's algorithm was applied by sorting all edges and selecting the minimum weighted edges using a union-find approach to form the Minimum Spanning Tree. Additionally, Prim's algorithm was used to grow the MST from a starting node using a greedy approach with minimum edge selection. These implementations together ensured efficient network traversal, connectivity analysis and cost-optimized infrastructure design.

Implementation Details:

1. Graph is represented using an adjacency list with weighted edges.
2. BFS is used to explore all nodes level-by-level using a queue.
3. DFS is implemented using recursion/stack for deep traversal and connectivity analysis.
4. Kruskal's algorithm sorts all edges and builds MST using Union-Find (Disjoint Set).
5. Prim's algorithm expands MST using a priority queue (min-heap) for greedy selection.
6. Final MST ensures minimum cost infrastructure design with $O(E \log E)$ or $O(E \log V)$ complexity.

Output:

```

import java.util.*;

class Edge {
    int u, v, w;

    Edge(int u, int v, int w) {
        this.u = u;
        this.v = v;
        this.w = w;
    }
}

class DSU {
    int[] parent;

    DSU(int n) {
        parent = new int[n];
        for (int i = 0; i < n; i++)
            parent[i] = i;
    }

    int find(int x) {
        if (parent[x] != x)
            parent[x] = find(parent[x]);
        return parent[x];
    }

    void union(int a, int b) {
        parent[find(a)] = find(b);
    }
}

public class SmartGrid {
    static void BFS(List<List<int[]>> g, int start) {
        boolean[] vis = new boolean[g.size()];
    }
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

● PS D:\dsa 2 case studies> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\balap\AppData\Roaming\Code\User\workspaceStorage\e494962941128a6f64546540eaa20d\redhat.java\jdt_ws\dsa 2 case studies_68bfcacc\bin' 'SmartGrid'
BFS: 0 1 2 3
DFS: 0 1 3 2
MST (Kruskal):
2-3 : 4
0-3 : 5
0-1 : 10
Total Cost = 19
○ PS D:\dsa 2 case studies>

```

SmartGrid demonstrates how graph algorithms optimize real-world infrastructure systems. BFS and DFS ensure efficient traversal and connectivity analysis, while Kruskal's and Prim's algorithms construct a Minimum Spanning Tree to minimize total construction cost. This combination enables optimal design of networks such as power grids, transportation systems, and communication networks with minimal cost and maximum efficiency.

Github Link: <https://github.com/yaswanthsurya-99/DSA-2>

Student Signature	Faculty Signature